

Bibliometric Baseline Analysis

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1 Summary

This report applies Bradford’s Law and Lotka’s Law to characterize the structure of subiculum research literature. I analyze 3,572 papers published across 592 journals from 1966–2025, identifying core sources, prolific contributors, and productivity patterns.

Key Findings:

- 12 core journals publish 34% of literature
- 83.3% of authors contribute only one paper (one-time contributors)
- 22 highly productive authors (11+ papers) represent core of field

2 Theoretical Background

2.1 Bradford’s Law

Bradford’s Law describes journal distribution following a $1:n:n^2$ pattern, where journals divide into three zones of equal paper output but exponentially increasing journal counts.

2.2 Lotka’s Law

Lotka’s Law states that author productivity follows $A_n = A_1/n^\alpha$ where $\alpha \approx 2$, meaning author counts decrease with the inverse square of publication number.

3 Bradford’s Law Results

3.1 Zone Distribution

Table 1: Bradford zone distribution

Zone	Journals	Papers	% of Total
Zone 1 (Core)	12	1,218	34.1%
Zone 2 (Middle)	62	1,197	33.5%
Zone 3 (Peripheral)	518	1,157	32.4%
Total	592	3,572	100%

Bradford multiplier:

- Zone 2 / Zone 1: $n_1 = 5.17$
- Zone 3 / Zone 2: $n_2 = 8.35$
- Average: $n \approx 6.76$

Bradford's Law: $1:n:n^2$ Pattern

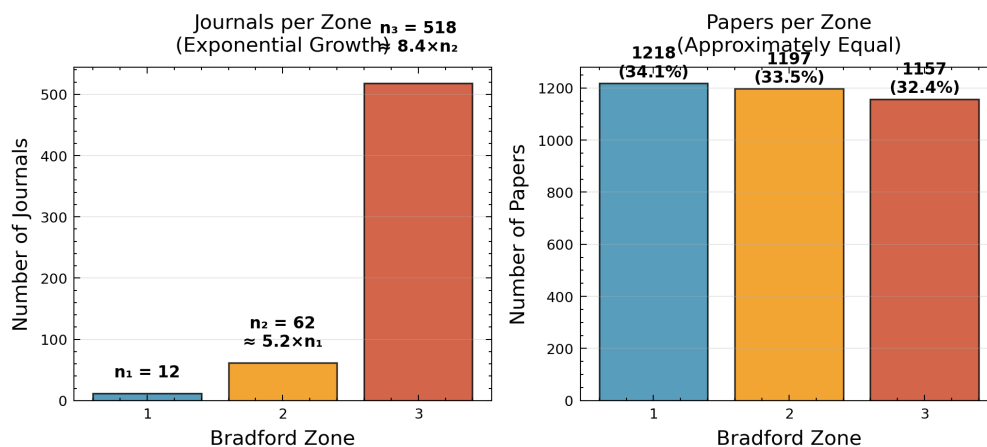


Figure 1: Bradford's Law $1:n:n^2$ pattern showing exponential journal growth (left) versus equal paper contributions (right).

Bradford Curve: Journal Productivity Distribution

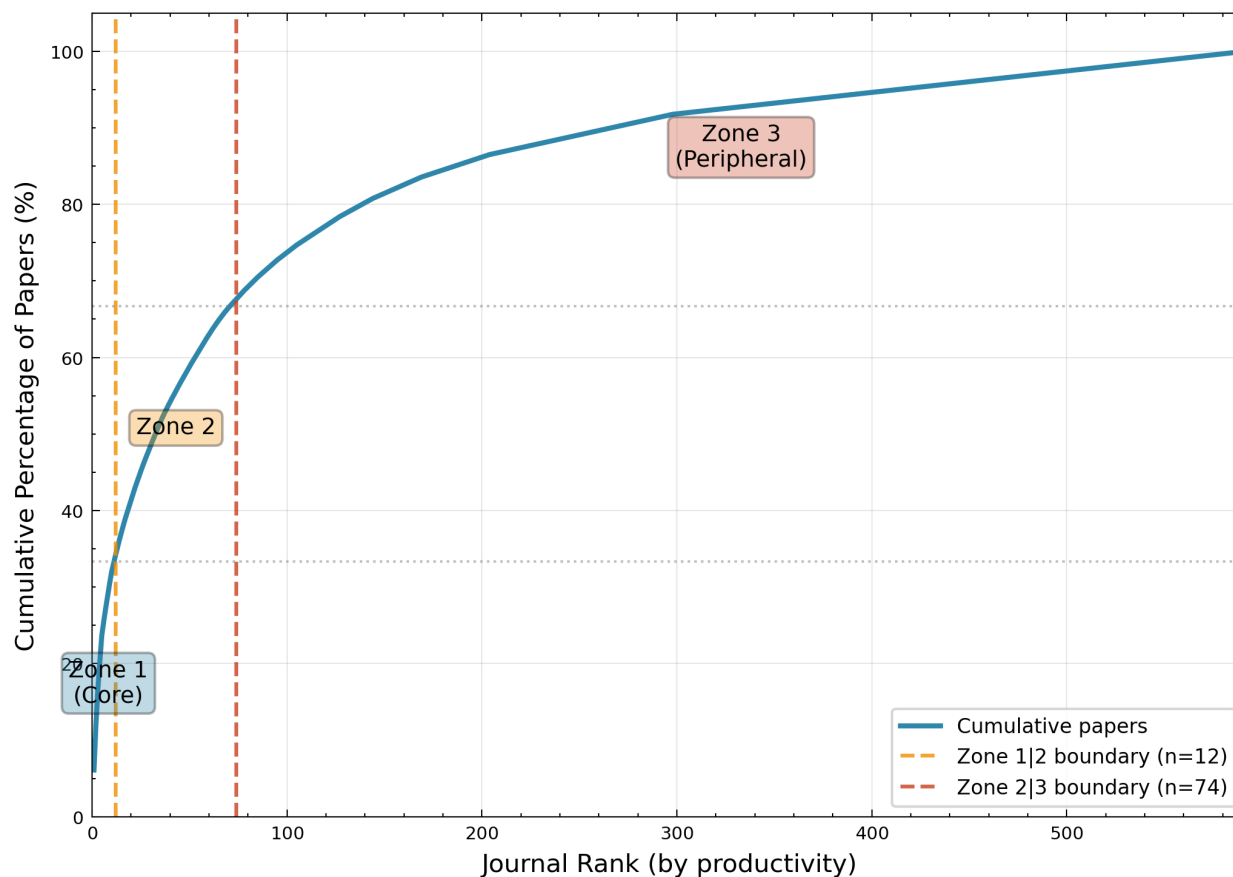


Figure 2: Bradford curve: cumulative paper percentage versus journal rank. Vertical lines mark zone boundaries.

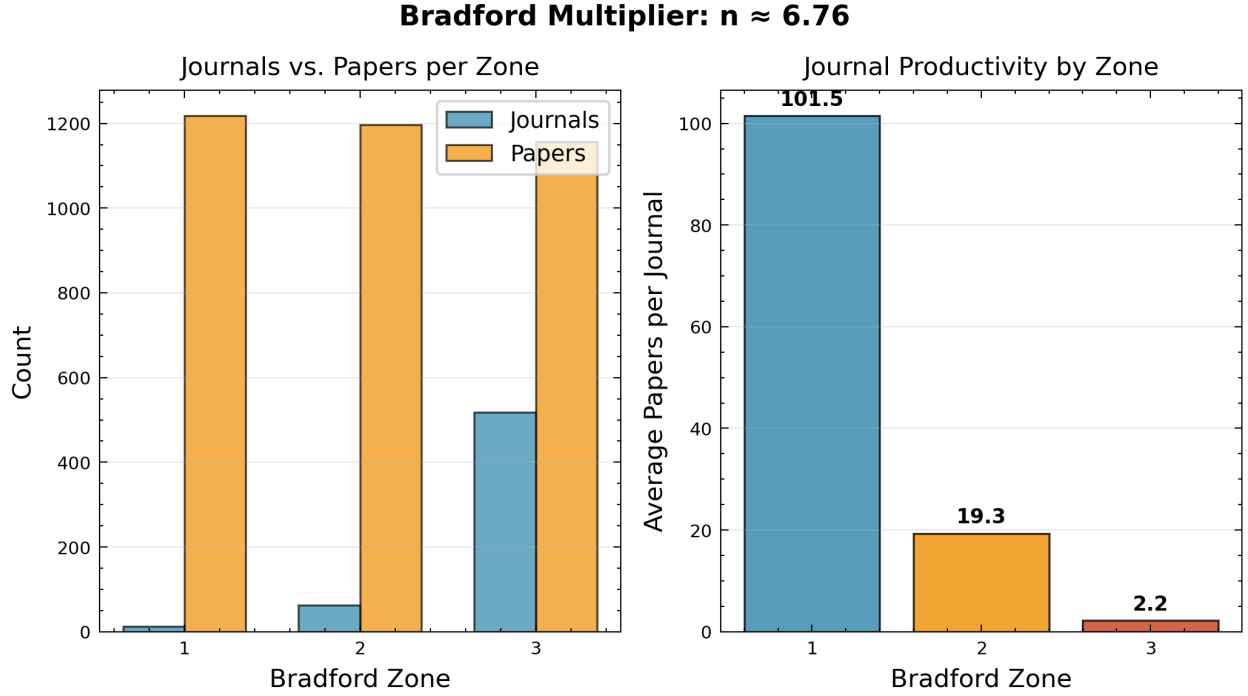


Figure 3: Bradford zone comparison showing journal/paper counts (left) and average productivity per zone (right).

3.2 Core Journals

The 12 core journals (Zone 1) provide access to one-third of subiculum literature. See Appendix A for complete list.

4 Lotka's Law Results

4.1 Power Law Fit

Table 2: Lotka's Law fit results

Parameter	Value
Fitted exponent (α)	3.12
Theoretical exponent	2.00
R^2	0.962
KS test p -value	0.888

The fitted $\alpha = 3.12$ deviates from theoretical 2.0 but remains within typical ranges for scientific literature (2–4). High R^2 (0.962) and non-significant KS test ($p = 0.888$) validate the power law model.

Empirical validation: The log-log plot (left panel, Figure 5) linearizes the power law relationship, where slope = $-\alpha$. Strong linear fit across productivity spectrum confirms power law behavior. The linear-scale comparison (right panel) shows model fit using first 10 productivity bins, demonstrating excellent agreement between observed author counts (points) and predicted values (line) for low-to-moderate productivity authors. Deviation at high productivity reflects natural sampling variation in small elite group.

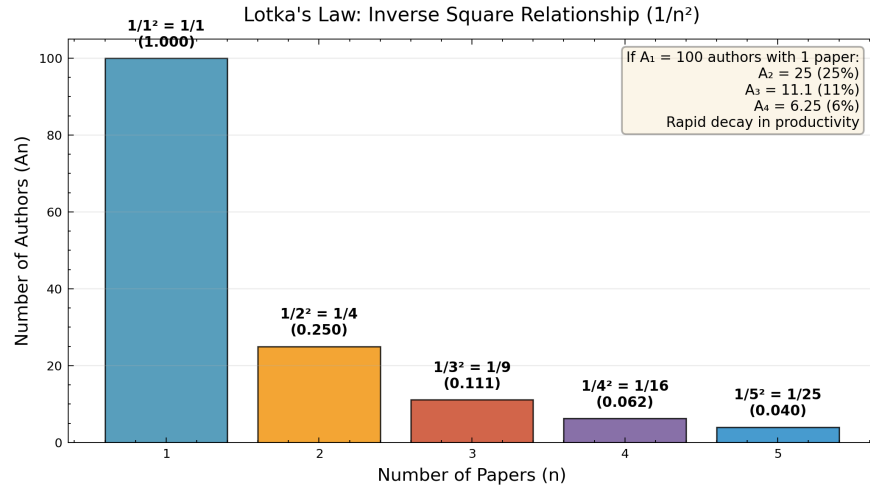


Figure 4: Lotka's Law conceptual illustration showing $1/n^2$ decay in author counts.

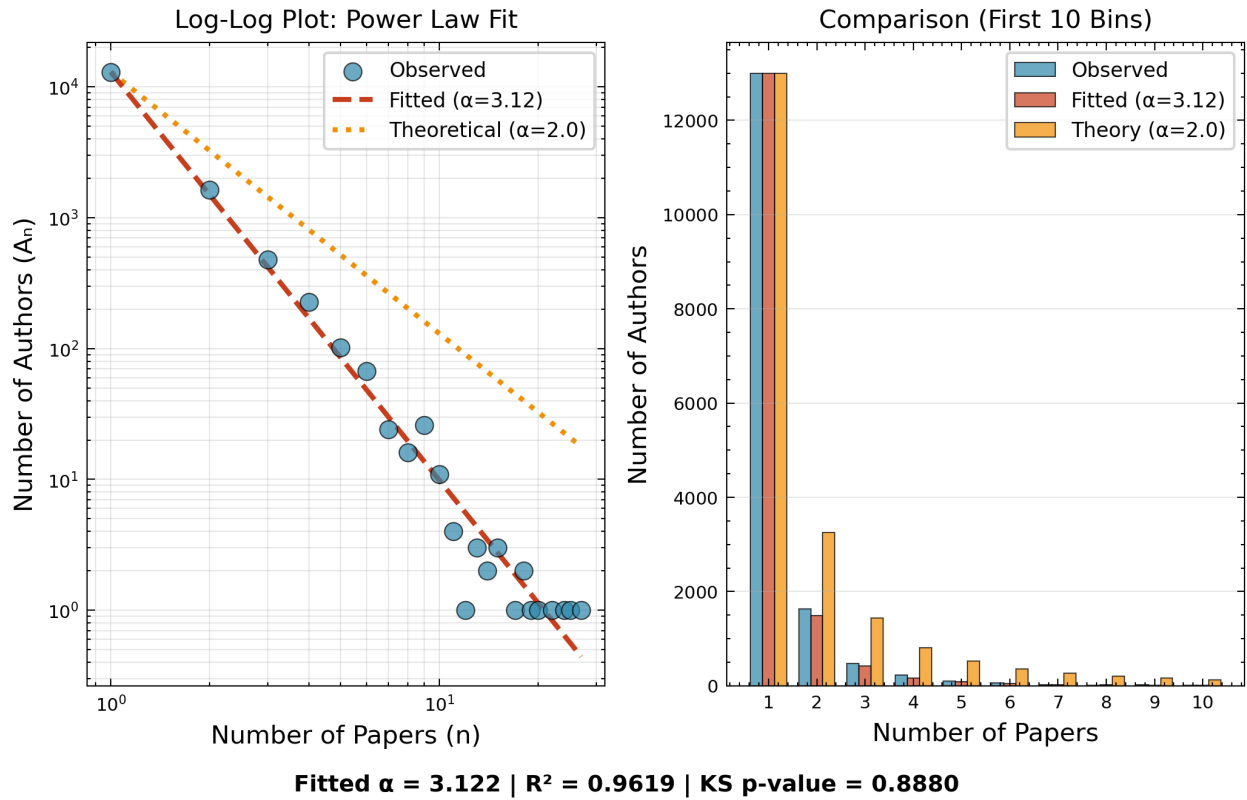


Figure 5: Empirical validation: observed data (points) versus fitted power law (lines) in log-log (left) and linear (right) scales.

4.2 Author Productivity Tiers

Table 3: Author productivity distribution

Tier	Range	Authors	% of Total
One-time	1 paper	12,996	83.3%
Occasional	2–5 papers	2,437	15.6%
Regular	6–10 papers	144	0.9%
Core	11+ papers	22	0.1%

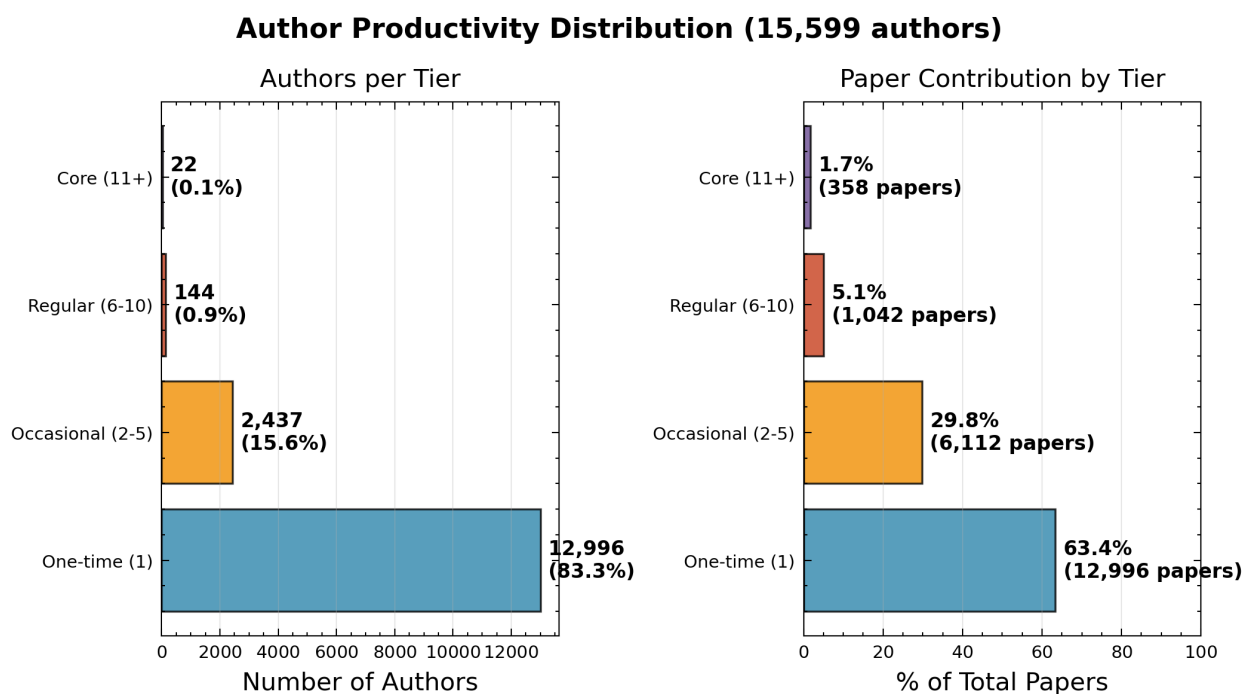


Figure 6: Author productivity tiers showing distribution (left) and paper contributions (right).

Interpretation: High concentration with 83.3% one-time contributors and 0.1% core contributors indicates typical scientific productivity patterns with small expert elite.

See Appendix B for top 25 most prolific authors.

5 Geographic Distribution

USA leads with 26.9% of papers, followed by Germany (6.5%), Japan (6.4%), and China (5.9%). Total of 23 countries represented. Complete data in Appendix C.

6 Funding Analysis

Summary:

- 42.2% of papers acknowledge funding
- 382 unique funding agencies

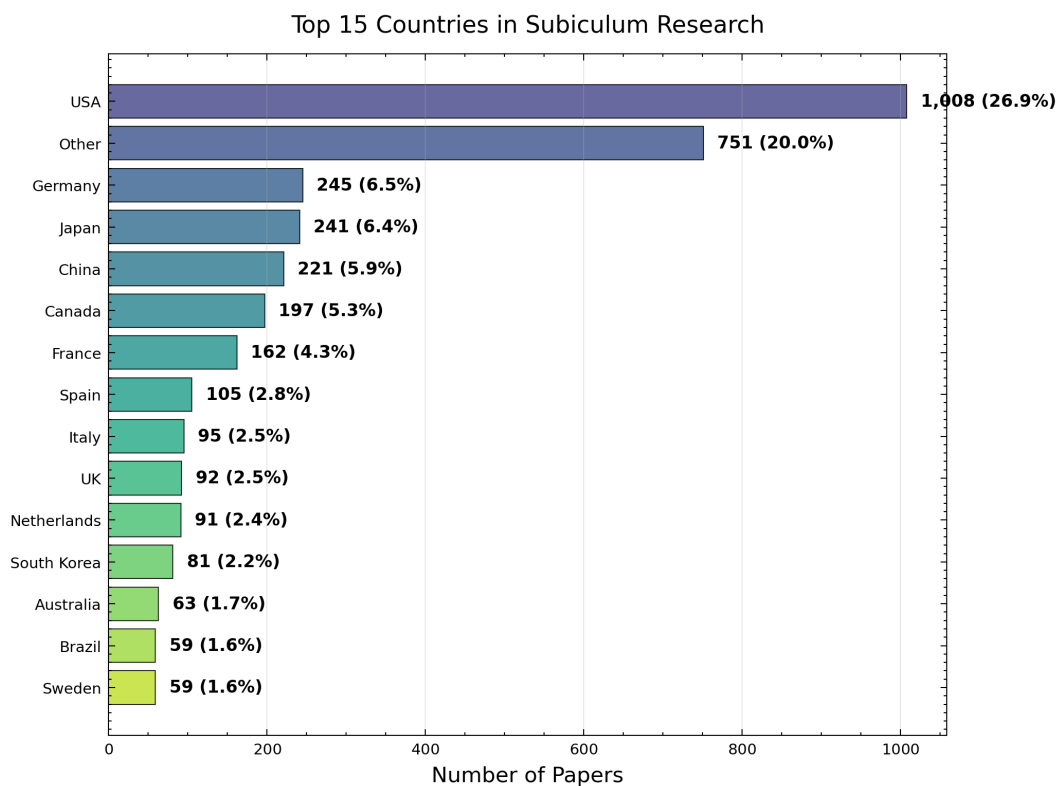


Figure 7: Top 15 countries by paper count showing USA dominance followed by European nations.

- Recent decade funding rate: 50.1%
- NIH dominates with 76.6% of funded papers

Complete funding data in Appendix D.

7 Conclusions

This bibliometric analysis establishes the quantitative baseline of subiculum research. Both Bradford's Law and Lotka's Law hold, indicating a mature field with typical concentration patterns:

- **Concentrated sources:** 12 core journals provide 34% coverage
- **Elite productivity:** Small group of 22 highly productive authors
- **International scope:** 23 countries with USA dominance (27%)
- **Strong funding:** 77% supported by NIH and European programs

For literature reviews: Focus on Zone 1–2 journals (74 total) for 68% coverage, supplemented by citation chasing.

For research strategy: Target core journals for publication; track work of 22 core contributors; pursue NIH/European funding.

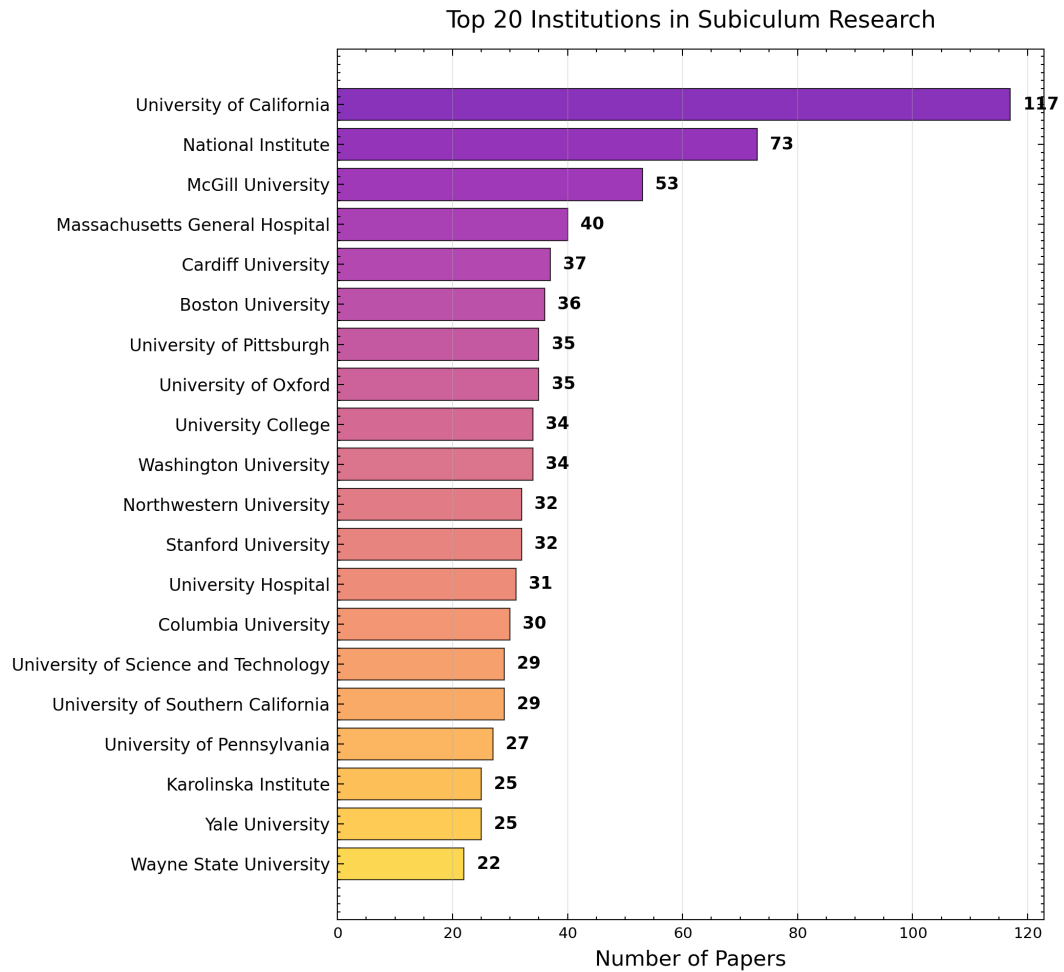


Figure 8: Top 20 institutions publishing subiculum research.

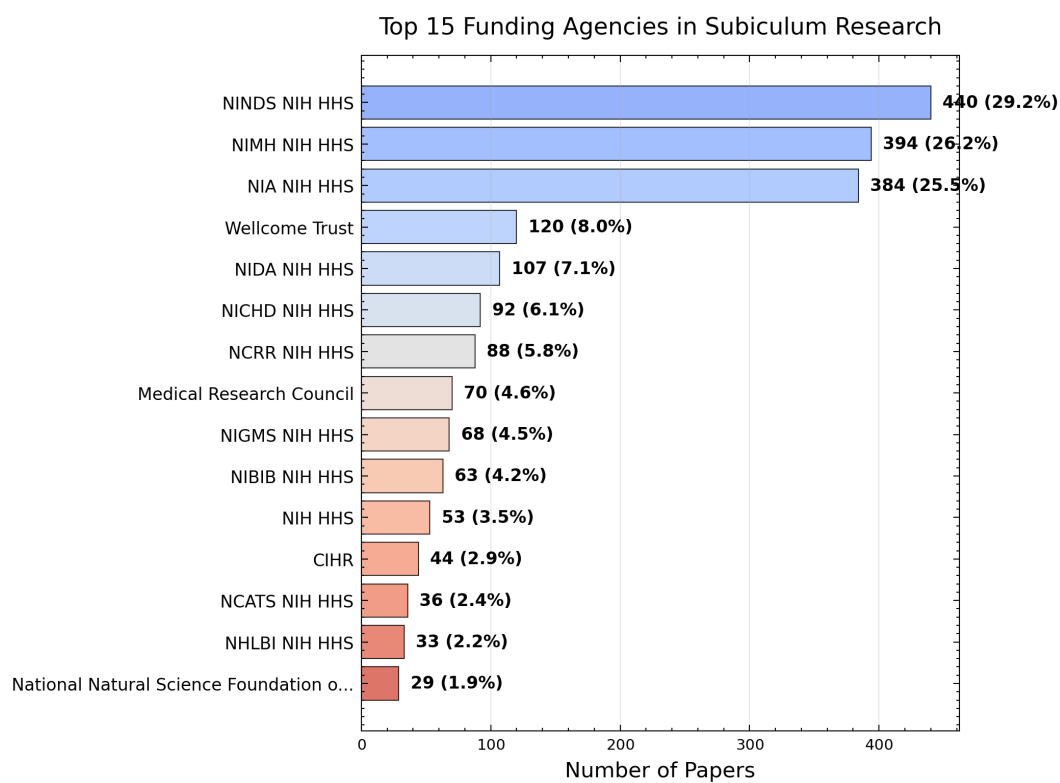


Figure 9: Top 15 funding agencies, dominated by NIH and European programs.

A Core Journals (Bradford Zone 1)

Table 4: Core journals publishing subiculum research (Zone 1)

Journal	Papers	Cumulative %
Brain research	218	6.1%
The Journal of comparative neurology	212	12.0%
Neuroscience	166	16.7%
Hippocampus	128	20.3%
The Journal of neuroscience	122	23.7%
The European journal of neuroscience	65	25.5%
Neuroscience letters	62	27.2%
Behavioural brain research	57	28.8%
Acta neuropathologica	57	30.4%
NeuroImage	54	31.9%
Brain research bulletin	39	33.0%
Scientific reports	38	34.1%

B Top 25 Authors

Table 5: Most prolific authors in subiculum research

Author	Papers	Career Span (yrs)	Tier
Aggleton, John P	27	22	Core (11+)
Witter, M P	25	38	Core (11+)
Heinemann, U	24	29	Core (11+)
Thompson, Paul M	22	20	Core (11+)
Grace, Anthony A	20	17	Core (11+)
Köhler, C	19	12	Core (11+)
Behr, Joachim	18	19	Core (11+)
Avoli, Massimo	18	19	Core (11+)
Dickson, Dennis W	17	19	Core (11+)
Witter, Menno P	15	20	Core (11+)
Avoli, M	15	14	Core (11+)
Swanson, L W	15	26	Core (11+)
O'Mara, Shane M	14	19	Core (11+)
Daugherty, Ana M	14	12	Core (11+)
Wang, Lei	13	22	Core (11+)
Heinemann, Uwe	13	16	Core (11+)
Amaral, D G	13	16	Core (11+)
Fidzinski, Pawel	12	12	Core (11+)
Jevtovic-Todorovic, Vesna	11	14	Core (11+)
Williams, Sylvain	11	8	Core (11+)
Behr, J	11	19	Core (11+)
Palacios, J M	11	12	Core (11+)
Wozny, Christian	10	20	Regular (6-10)
Raz, Naftali	10	14	Regular (6-10)
Wang, Yi	10	13	Regular (6-10)

C Geographic Distribution

C.1 Country Distribution

Table 6: Papers by country

Country	Papers	% of Total
USA	1,008	26.9%
Other	751	20.0%
Germany	245	6.5%
Japan	241	6.4%
China	221	5.9%
Canada	197	5.3%
France	162	4.3%
Spain	105	2.8%
Italy	95	2.5%
UK	92	2.5%
Netherlands	91	2.4%
South Korea	81	2.2%
Australia	63	1.7%
Brazil	59	1.6%
Sweden	59	1.6%
Switzerland	56	1.5%
Norway	39	1.0%
Denmark	39	1.0%
Austria	38	1.0%
India	33	0.9%
Finland	26	0.7%
Israel	25	0.7%
Belgium	23	0.6%

C.2 Top Institutions

Table 7: Top 30 institutions publishing subiculum research

Institution	Papers
University of California	117
National Institute	73
McGill University	53
Massachusetts General Hospital	40
Cardiff University	37
Boston University	36
University of Pittsburgh	35
University of Oxford	35

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Table 7 – *Continued from previous page*

Institution	Papers
University College	34
Washington University	34
Northwestern University	32
Stanford University	32
University Hospital	31
Columbia University	30
University of Science and Technology	29
University of Southern California	29
University of Pennsylvania	27
Karolinska Institute	25
Yale University	25
Wayne State University	22
Zhejiang University	20
University of Toronto	20
University of Medical Sciences	19
University of British Columbia	19
Tokyo Metropolitan Institute	18
Medical University	18
University of Cambridge	18
University Medical Center	18
Douglas Mental Health University	18

D Funding Agencies

Table 8: Top 30 funding agencies supporting subiculum research

Agency	Papers	% of Funded
NINDS NIH HHS	440	29.2%
NIMH NIH HHS	394	26.2%
NIA NIH HHS	384	25.5%
Wellcome Trust	120	8.0%
NIDA NIH HHS	107	7.1%
NICHD NIH HHS	92	6.1%
NCRR NIH HHS	88	5.8%
Medical Research Council	70	4.6%
NIGMS NIH HHS	68	4.5%
NIBIB NIH HHS	63	4.2%
NIH HHS	53	3.5%
CIHR	44	2.9%
NCATS NIH HHS	36	2.4%
NHLBI NIH HHS	33	2.2%
National Natural Science Foundation of China	29	1.9%
NIAAA NIH HHS	27	1.8%
Intramural NIH HHS	27	1.8%
NIDDK NIH HHS	26	1.7%
NCI NIH HHS	23	1.5%
Canadian Institutes of Health Research	21	1.4%
PHS HHS	19	1.3%
NEI NIH HHS	17	1.1%
NIDCD NIH HHS	14	0.9%
Biotechnology and Biological Sciences Research Council	13	0.9%
Howard Hughes Medical Institute	11	0.7%
NIDCR NIH HHS	10	0.7%
Austrian Science Fund FWF	9	0.6%
NLM NIH HHS	9	0.6%
CSRD VA	8	0.5%
Deutsche Forschungsgemeinschaft	8	0.5%